

Machine Learning Project Report

Team Member:

Phạm Minh Hiếu – 523V0001

Hồ Lê Minh – 523K0015

1. Introduction

The objective of this project is to design, implement, and evaluate a complete machine learning pipeline to predict whether a student will receive a scholarship based on their academic performance and socio-economic background. Scholarship allocation is a critical decision-making process in educational institutions, requiring both high accuracy to ensure fairness and high recall to ensure no deserving student is overlooked. By leveraging a synthetic tabular dataset, this project explores exploratory data analysis (EDA), feature engineering, data preprocessing, and the comparative evaluation of classical machine learning algorithms, specifically Logistic Regression and Random Forest. The ultimate goal is to identify a robust model capable of generalizing well to unseen data while providing interpretable results.

2. Dataset

The dataset utilized in this project consists of synthetic student records from a hypothetical country, specifically designed for educational machine learning purposes. It poses a binary classification problem.

The training dataset contains 200 individual student records with no missing values. The features are defined as follows:

- `id`: A unique identifier for each student (integer).
- `gpa`: The student's Grade Point Average, ranging from 2.01 to 3.97 (float).
- `attendance_rate`: The proportion of classes attended, ranging from 0.60 to 1.00 (float).
- `study_hours_per_week`: The amount of self-study time per week, ranging from 5 to 34 hours (integer).
- `exam_score`: The final examination score, ranging from 50 to 99 (integer).
- `household_income`: The annual household income of the student's family in USD, ranging from \$200 to \$1999 (integer).
- `label` (Target Variable): A binary indicator where 1 denotes the student received a scholarship, and 0 denotes they did not.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the underlying distributions and relationships between features and the target variable. Descriptive statistics confirmed that the dataset is clean, with no missing values (Null count = 0 across all features), eliminating the need for imputation.

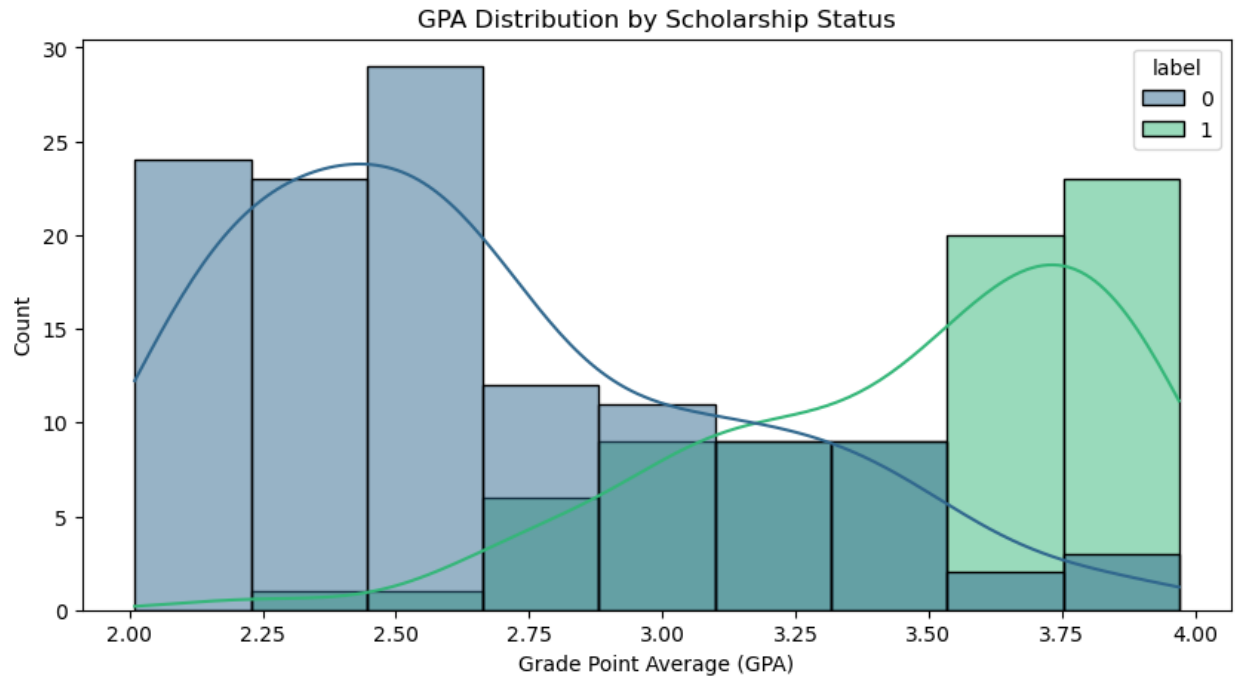


Figure 1. Distribution of Grade Point Average (GPA) segmented by scholarship receipt status.

It is evident that students receiving the scholarship (Label 1) are heavily clustered at the higher end of the GPA spectrum (3.5 to 4.0), indicating that academic excellence is a primary driver for the award.

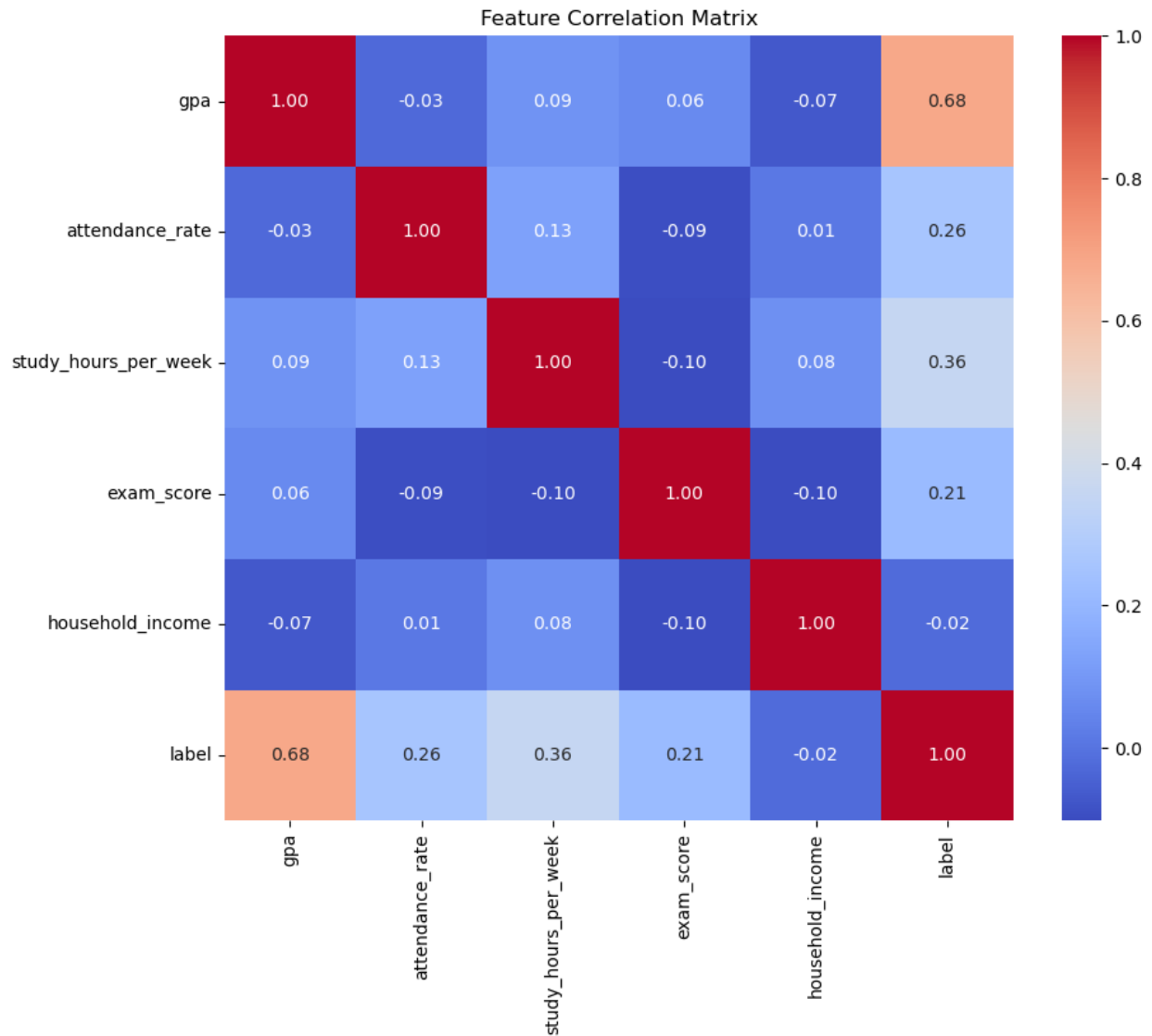


Figure 2. Feature correlation matrix heatmap demonstrating the linear relationships between variables and the target label.

A correlation matrix was generated to quantify the linear relationships between features. The matrix reveals that gpa exhibits the strongest positive correlation with the target label ($r=0.68$), followed by study_hours_per_week ($r=0.36$) and attendance_rate ($r=0.26$). Interestingly, household_income demonstrates a near-zero linear correlation with the label ($r=-0.02$).

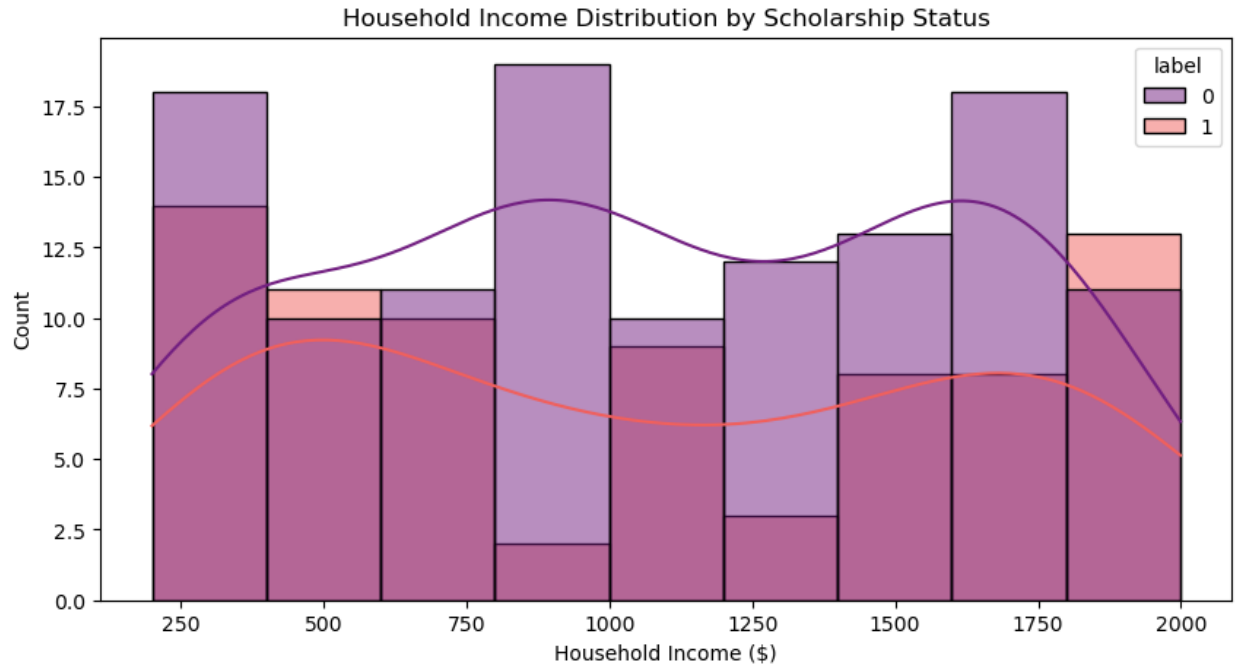


Figure 3. Histogram of annual household income illustrating a non-linear threshold effect on scholarship allocation.

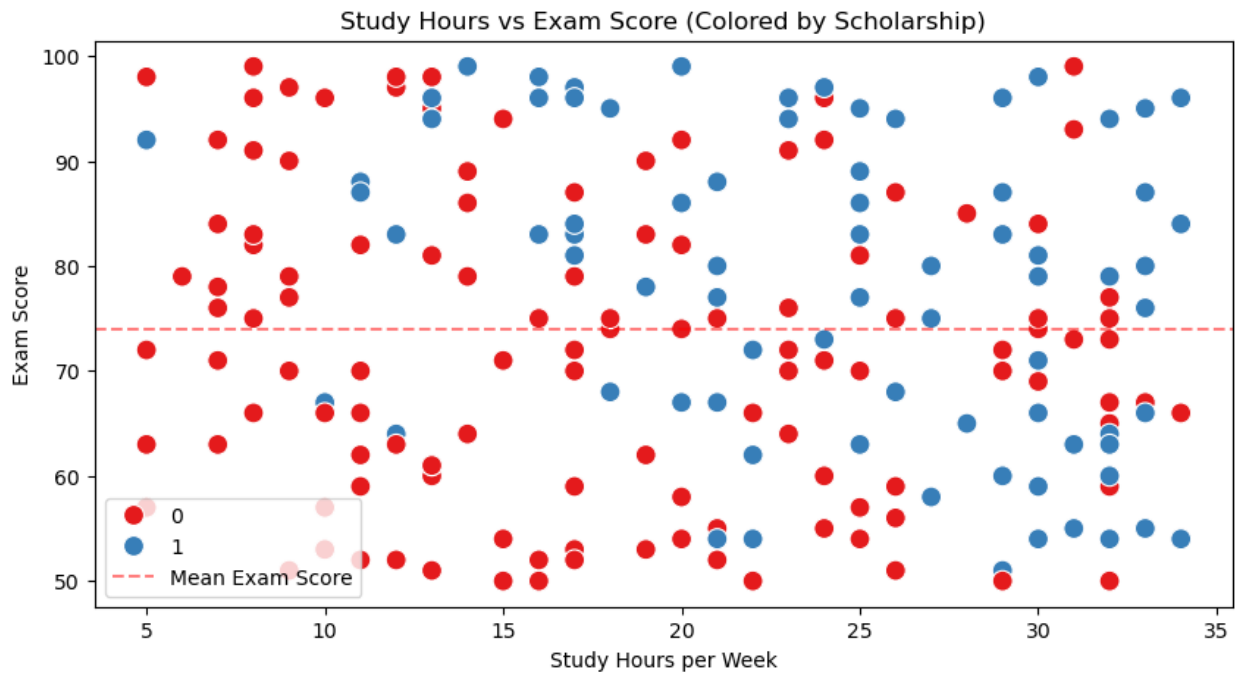


Figure 4. Scatter plot of study hours versus exam scores, highlighting the distribution of scholarship recipients relative to the mean exam score.

Despite the low linear correlation, the distribution of household_income (Figure 3) and the scatter plot of study_hours vs exam_score (Figure 4) suggest that income may act as a hard

threshold rather than a linear predictor. High academic achievers are frequently denied scholarships if their household income exceeds a certain socio-economic threshold.

4. Data Preprocessing

To optimize the dataset for machine learning algorithms, the following preprocessing and feature engineering steps were executed:

4.1 Feature Engineering

To capture the synergistic effects of a student's attributes, two new interaction terms were engineered:

1. `gpa_attendance`: The product of `gpa` and `attendance_rate`. This represents a combined metric of academic capability and diligence.
2. `study_exam`: The product of `study_hours_per_week` and `exam_score`. This captures the relationship between effort and actual academic outcome.

4.2 Feature Scaling and Justification

The numerical features in this dataset possess vastly different scales (e.g., `gpa` maxes at 4.0, while `household_income` reaches up to 2000). To address this, **StandardScaler** was applied to normalize the feature space to a mean of 0 and a variance of 1.

Justification: Standardization is strictly required for gradient-based models like Logistic Regression. Without scaling, features with larger magnitudes (like income) would disproportionately dominate the cost function, leading to biased weight updates and poor convergence. Standardizing ensures that the model evaluates the relative importance of each feature purely based on its variance and relationship with the target, rather than its raw mathematical magnitude.

5. Models

Two distinct supervised learning algorithms were trained to establish a comparative baseline: a linear model and a non-linear ensemble model.

5.1 Logistic Regression (Baseline)

Logistic Regression was selected as the baseline model due to its high interpretability, efficiency, and strong performance on linearly separable data. To ensure optimal performance, hyperparameter tuning was conducted using a 5-fold Grid Search Cross-Validation (`GridSearchCV`). The regularization parameter `C` was tuned across values `[0.1, 1, 10, 50]`. The `liblinear` solver was utilized with a `max_iter` of 2000 to guarantee convergence.

5.2 Random Forest (Advanced)

To account for potential non-linear relationships and complex decision boundaries (such as income thresholds), a Random Forest Classifier was implemented. This ensemble method builds multiple decision trees to reduce variance and prevent overfitting. Hyperparameters were similarly tuned via 5-fold GridSearchCV, specifically exploring the number of trees (n_estimators: 200, 300) and the maximum depth of the trees (max_depth: 10, 15, None).

6. Experimental Results

The models were evaluated on the Development (Dev) set using Accuracy, Precision, Recall, and the F1-Score. For this specific educational context, **Recall** is heavily prioritized, as failing to grant a scholarship to a deserving student (False Negative) is deemed highly undesirable.

Table 1: Model Performance Comparison

Metric	Logistic Regression	Random Forest
Accuracy	0.9750	0.9500
Precision	0.9412	0.9375
Recall	1.0000	0.9375
F1 Score	0.9697	0.9375

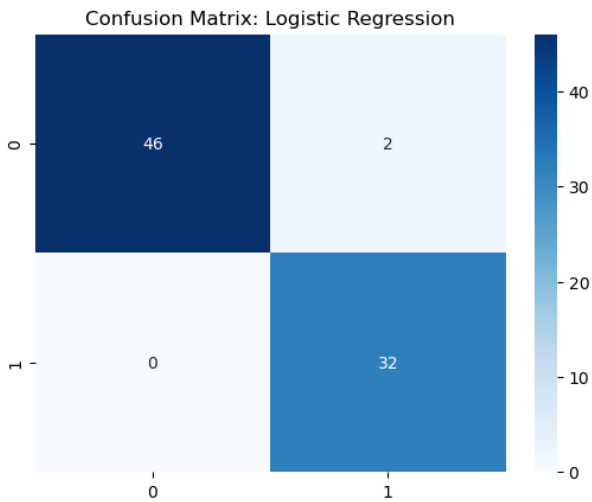


Figure 5. Confusion matrix for the Logistic Regression baseline model evaluated on the Development set.

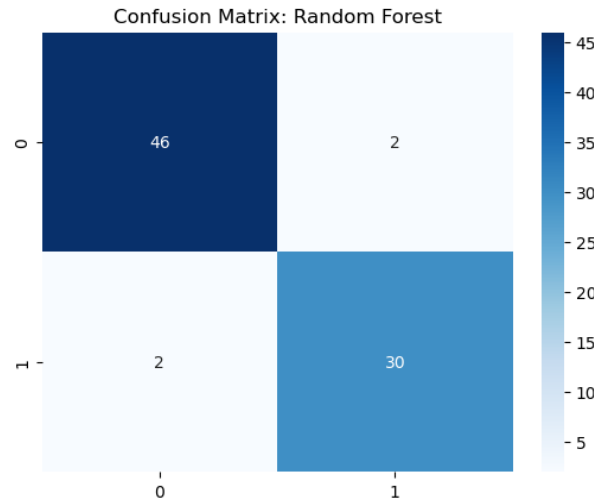


Figure 6. Confusion matrix for the Random Forest ensemble model evaluated on the Development set.

The Logistic Regression model outperformed the Random Forest model across all metrics. Notably, it achieved a perfect Recall of 1.0000 (0 False Negatives), meaning every deserving student was successfully identified. The Confusion Matrix for Logistic Regression reveals only 2 False Positives out of the entire evaluation set.

7. Error Analysis

An in-depth analysis was conducted on the 2 misclassified samples (False Positives) generated by the Logistic Regression model on the Dev set. In these cases, the model predicted that the student should receive a scholarship, but the ground truth was negative.

1. **Sample ID 18:** This student exhibited an exceptional exam_score of 96 and invested 21 study_hours_per_week.
2. **Sample ID 67:** This student maintained an outstanding gpa of 3.58 and an attendance_rate of 0.94.

Discussion of Causes: In both instances, the model was heavily influenced by the students' stellar academic metrics, which strongly correlate with positive labels. However, both students possessed extremely high household_income values (\$1742 and \$1673, respectively). This suggests that the scholarship granting mechanism employs a strict "hard threshold" (e.g., automatic rejection if income > \$1600), regardless of academic brilliance. Linear models like Logistic Regression apply continuous penalization and struggle to replicate sudden hard-threshold cutoffs, resulting in these specific False Positives.

8. Conclusion

In this project, a comprehensive machine learning pipeline was successfully developed to predict scholarship eligibility. Through rigorous EDA, feature engineering, and robust preprocessing, the data was adequately prepared for modeling.

The experimental results demonstrated that Logistic Regression is highly effective for this dataset, achieving an F1-Score of 0.9697 and a flawless Recall of 1.0. However, the error analysis highlighted a critical limitation: linear models struggle with socio-economic hard thresholds.

Potential Improvements: In future iterations, this limitation could be mitigated by applying "binning" or discretization to the household_income feature (e.g., categorizing into Low, Medium, High income tiers) to help linear models comprehend the threshold. Alternatively, implementing advanced gradient boosting frameworks such as XGBoost or LightGBM could natively capture these non-linear, tree-based splits, potentially eliminating the remaining False Positives and achieving perfect classification accuracy.